The pair singular series of a polynomial III: the off-diagonal in Cesàro form

Jasper Reichardt Claude Fable 5.1

Draft of September 10, 2026

Abstract

For an irreducible quadratic f with pair singular series $S_f(h)$ and Bateman–Horn constant $C(f)$, parts I and II reduced the conjecture $\sum_{h\le H}(S_f(h)-C(f)^2) = -\frac{1}{2}C(f)\log H + A_f + o(1)$ to a statement about the pairs of distinct roots of f modulo squarefree d ("Hypothesis (E)"), and showed that in Cesàro form any unbounded saving over the trivial bound suffices. This paper takes Hypothesis (E) apart. We prove an exact decomposition of the Cesàro off-diagonal into pieces indexed by the part u of the modulus on which the two roots agree: piece u is the same problem with length H/u and with the roots of $x^2 \equiv D$ replaced by the roots of $u^2x^2 \equiv D$, a dilation by u^{-1}, and all pieces carry equal weight per unit of $\log u$. For $u = 1$ the required equidistribution is the theorem of Duke, Friedlander and Iwaniec on Weyl sums for quadratic roots, which we convert from smooth to sharp cut-offs; for $u > H^{1/2}$ the dilations average out along u and only the distribution of a divisor-type function in arithmetic progressions is needed; for $1 < u \le H^{1/2}$ the dilated Weyl sums are the open input, stated as Hypothesis W with the exponent it must have. Under Hypothesis W the Cesàro form of Hypothesis (E) holds for every quadratic. Over $\mathbb{F}_q[u]$ we prove that the off-diagonal is not merely bounded but has a main term: for $f = t^2 - u$ and the two smallest lengths, $\mathrm{Off}_f(N) = -2/q + O(q^{-3/2})$; a configuration count predicts, and exact computations for three discriminants and $q \le 43$ confirm, that the generic value is $-1/q$, the weight of the excluded diagonal pairs at the first degree above the cut-off, the value $-2/q$ being a small-length effect.

1 Introduction

In plain terms. Prime values of a polynomial are correlated through the small primes. Part I turned the size of that correlation into an exact identity whose main part, coming from equal roots of f modulo d, is understood, and whose remaining part, coming from pairs of distinct roots, is a question about how the square roots of the discriminant modulo composite numbers are distributed. This paper shows that the remaining part is a sum of copies of one problem, one copy for each way the two roots can agree on a part of the modulus, that the copies are of equal importance, that the first copy is a theorem of Duke, Friedlander and Iwaniec, that the copies with a large agreeing part average out, and that the copies in between require one estimate that we state precisely but do not prove. Over polynomial rings the same object has an explicit main term, which we compute.

Set-up. Let $f = t^2 + bt + c \in \mathbb{Z}[t]$ be irreducible, without fixed prime divisor, with discriminant $D = b^2 - 4c$ (not a square). For a prime p let $\omega(p) = \omega_f(p)$ be the number of roots of f modulo p: $\omega(p) = 2$ if $p \nmid 2D$ and D is a square modulo p (*split*), 0 if D is a non-square (*inert*), and ≤ 1 if $p \mid 2D$ (*special*). All moduli below are squarefree. Put

$$\lambda(p) = \frac{p}{p - 2\omega(p)}, \qquad P_p = 1 - \frac{\omega(p)^2}{(p-\omega(p))^2}, \qquad P(1) = \prod_p P_p, \qquad W_f(d) = \frac{P(1)\lambda(d)}{d},$$

λ multiplicative; this is the weight of part I, $W_f(d) = d\,b(d)\prod_{p\nmid d} P_p$ with $b(p) = (p-\omega(p))^{-2}$, since $p\,b(p)/P_p = 1/(p-2\omega(p))$. For d coprime to $2D$ let $R_d = \{r \bmod d : \; r^2 \equiv D\}$ (so $|R_d| = \omega(d)$), the roots of f being $s = (r-b)/2$. Write $\langle x\rangle_d \in [1,d]$ for the representative of x mod d, and, for a length $Y > 0$ and a residue m,

$$B_d^{(Y)}(m) = \sum_{\substack{h\le Y\\ h\equiv m\ (d)}} (Y-h) - \frac{Y^2}{2d}. \tag{1}$$

Part I, Theorem 6, proved that with $C = C(f)$

$$\sum_{h\le H}\Big(1-\frac{h}{H}\Big)\big(S_f(h)-C^2\big) = -\tfrac12 C\log H + O_f(1) + \frac{C^2}{H}\mathrm{Off}^*_f(H), \qquad \mathrm{Off}^*_f(H) = \sum_d W_f(d) \sum_{\substack{s,s'\ \text{roots of } f \bmod d\\ s\neq s'}} B_d^{(H)}(\tag{2}$$

the series converging absolutely, and that $\mathrm{Off}^*_f(H) \ll H\log H$ trivially. *Hypothesis (E) in Cesàro form* is $\mathrm{Off}^*_f(H) = o(H\log H)$; it implies Conjecture 1 of part I in Cesàro form with the leading term $-\frac12 C(f)\log H$ (part I, Theorem 6(ii) and Remark 7, since the Galois group S_2 is 2-transitive).

Results.

- *Decomposition* (Proposition 1): $\mathrm{Off}^*_f(H) = P(1)\sum_u \lambda(u)\omega(u)\,\mathcal{P}_u(H/u)$, the sum over squarefree u with $\omega(u)\ge 1$, where the piece $\mathcal{P}_u(Y)$ is the sum (2) with H replaced by Y and the roots $R_{d'}$ replaced by the roots $R^{(u)}_{d'}$ of $u^2x^2 \equiv D$, over moduli d' coprime to u all of whose primes are split. Each piece is $\ll Y$, and $\sum_u \lambda(u)\omega(u)H/u \asymp H\log H$.
- *The first piece* (Theorem 5): $\mathcal{P}_1(Y) \ll Y^{1-\delta}$ unconditionally when D is a fundamental discriminant covered by [2]; the input is a sharp-cut-off form of their Weyl-sum bound (Lemma 4).
- *Large agreeing part* (Theorem 7): $\sum_{u>H^{1/2+\varepsilon}} \lambda(u)\omega(u)\mathcal{P}_u(H/u) \ll H^{1-\delta}$, by averaging the dilations along u; the input is the distribution of $\lambda\omega$ in arithmetic progressions.
- *The remaining pieces* need Hypothesis W (§5): the Weyl sums of the dilated roots $R^{(u)}_{d'}$, with sharp cut-off in d', save a power of the range with a loss at most u^B, $B < 1-\theta$. Theorem 9: under Hypothesis W, $\mathrm{Off}^*_f(H) \ll H^{1-\delta}$ for every quadratic f with D as above, hence Conjecture 1 of part I in Cesàro form with a power-saving error.
- *Function field* (Theorem 10): for $f = t^2 - u$ over $\mathbb{F}_q[u]$ and $N \in \{1,2\}$, $\mathrm{Off}_f(N) = -2/q + O(q^{-3/2})$; Conjecture 13: $\mathrm{Off}_f(N) = -1/q + O(q^{-3/2})$ whenever $\max(2N-2, \deg D) \ge N+1$, i.e. for all $N \ge 3$ and for $N \le 2$ when $\deg D > N$, and $-2/q$ otherwise.

What is new relative to Hooley. The decomposition into pieces is implicit in Hooley's treatment of $\sum_{n\le x}\tau(n^2+a)$ [3], whose sum Σ_5 is our Off with the weight $1/k$ and whose Lemma 5 handles the phase that our piece $u = 1$ carries; what is new is the identification of the pieces $u > 1$ as dilated copies, their equal weight, the averaging argument for large u, the precise statement of what remains, and the function-field main term.

2 The exact decomposition

For squarefree d coprime to $2D$, the roots of f modulo d are $s = (r-b)/2$, $r \in R_d$, and $R_d = \prod_{p\mid d} R_p$ by the Chinese remainder theorem, each R_p being $\{\pm r_p\}$. Two roots $s \neq s'$

therefore agree exactly on a divisor $u \mid d$ and differ exactly on $d' = d/u > 1$, where every prime of d' is split; and on d' we have $r' \equiv -r$, so $s' - s \equiv -r \pmod{d'}$ and $s' - s \equiv 0 \pmod{u}$. Let $\bar{u}$ denote the inverse of u modulo d'. Then $s' - s \equiv u\bar{u} \cdot (-r) \pmod{d}$, i.e.

$$\langle s' - s \rangle_d = u \, \langle -\bar{u} r \rangle_{d'}. \tag{3}$$

If d has a special prime p ($\omega(p) = 1$) the two roots agree at p, so $p \mid u$; inert primes do not divide d at all ($\omega(d) = 0$). Define, for squarefree u with $\omega(u) \geq 1$ and squarefree d' coprime to u with all primes split,

$$R_{d'}^{(u)} = \{-\bar{u} r \bmod d' : \ r \in R_{d'}\} = \{x \bmod d' : \ u^2 x^2 \equiv D \ (d')\},$$

the roots of the fixed quadratic $Q_u(x) = u^2x^2 - D$ modulo d' (the equality because $x \mapsto -\bar{u}x$ is a bijection $R_{d'} \to R_{d'}^{(u)}$ when $(u, d') = 1$), and the *piece*

$$\mathcal{P}_u(Y) = \sum_{d'} \frac{\lambda(d')}{d'} \sum_{x \in R_{d'}^{(u)}} B_{d'}^{(Y)}\big(\langle x \rangle_{d'}\big). \tag{4}$$

Proposition 1 (pieces)**.** *For every $H \geq 1$,*

$$\mathrm{Off}_f^*(H) = P(1) \sum_u \lambda(u) \, \omega(u) \, \mathcal{P}_u(H/u),$$

the sum over squarefree $u \geq 1$ with $\omega(u) \geq 1$, every series converging absolutely. Moreover $\mathcal{P}_u(Y) \ll_f Y$ uniformly in u, and $\sum_{u \leq H} \lambda(u)\omega(u)/u \asymp \log H$.

Proof. Fix $d = ud'$ as above. For $m = u\mu$ with $\mu \in [1, d']$ we have, writing $h = uh'$,

$$B_{ud'}^{(H)}(u\mu) = \sum_{\substack{h' \leq H/u \\ h' \equiv \mu \ (d')}} (H - uh') - \frac{H^2}{2ud'} = u \, B_{d'}^{(H/u)}(\mu).$$

The ordered pairs (s, s') of distinct roots modulo d with agreeing part u are in bijection with pairs (s, d'), and by (3) their difference is $u\langle -\bar{u} r \rangle_{d'}$ where r is the d'-component of $2s + b$; as s runs over the $\omega(u)\omega(d')$ roots, r runs $\omega(u)$ times over $R_{d'}$. Hence

$$\sum_{s \neq s'} B_d^{(H)}(s' - s) = \sum_{\substack{d' \mid d, \ d' > 1 \\ d' \text{ split}}} \omega(u) \, u \sum_{r \in R_{d'}} B_{d'}^{(H/u)}\big(\langle -\bar{u} r \rangle_{d'}\big),$$

and with $W_f(ud') = P(1)\lambda(u)\lambda(d')/(ud')$ the claimed formula follows by summing over d; absolute convergence is inherited from (2), all terms being regrouped without change of absolute value. For the bound on $\mathcal{P}_u(Y)$: if $d' \leq Y$ then $|B_{d'}^{(Y)}(m)| \leq d'$ (part I, proof of Theorem 6), giving $\sum_{d' \leq Y} \lambda(d')\omega(d') \ll Y$ by Shiu's theorem, since $\lambda\omega$ is multiplicative with $\lambda\omega(p) \leq 2p/(p-4)$ on split primes and 0 on inert ones, of mean value $\asymp 1$. If $d' > Y$ then $B_{d'}^{(Y)}(m) = (Y - m)^+ - Y^2/(2d')$ for $m \in [1, d']$; the second term contributes $\ll Y^2 \sum_{d' > Y} \lambda(d')\omega(d')/d'^2 \ll Y$, and the first contributes $\sum_{x \leq Y} (Y - x) \sum_{d' > Y, \ d' \mid Q_u(x)} \lambda(d')/d'$; since $\sum_{d' > Y, \ d' \mid n} 1/d' \leq n^{-1} \sum_{e \mid n, \ e < n/Y} e$, and on average over $x \leq Y$, $\sum_{e < n/Y, \ e \mid Q_u(x)} e \leq \sum_{e < u^2 Y} e \, \rho_u(e) \, \mathbf{1}_{e \mid Q_u(x)}$ has mean $\ll \sum_{e < u^2 Y} \rho_u(e) \ll u^2 Y$ ($\rho_u(e) \leq \omega(e)$ the number of roots of Q_u modulo e, of mean value $\ll 1$), while $Q_u(x) \gg u^2 x^2$, the first term is $\ll \sum_{x \leq Y} (Y - x) \, Y/x^2 \cdot \mathbf{1}_{Q_u(x) > Y} + \sum_{x \leq \sqrt{Y}/u} Y \ll Y$. Finally $\sum_{u \leq H} \lambda(u)\omega(u)/u \asymp \log H$ because $\lambda\omega$ has mean value $\asymp 1$ on squarefree integers (its Dirichlet series is $\zeta(s) L(s, \chi_D) \times$ an absolutely convergent product near $s = 1$). ☐

In words. The off-diagonal is a sum over u of the same object, evaluated at length H/u, for the roots of $u^2x^2 \equiv D$ instead of $x^2 \equiv D$; since $\sum_u \lambda\omega(u)/u$ grows like $\log H$ while each piece is $O(H/u)$, the pieces are equally important on a logarithmic scale, and no range of u can be discarded. The dilation by $\bar{u}$ is the whole difficulty: for $u = 1$ the roots are those of a fixed fundamental quadratic; for $u > 1$ they are those of a quadratic whose discriminant $4u^2 D$ grows with u, and existing theorems degrade with the discriminant.

3 Reduction of a piece to Weyl sums

Fix u and write Y for the length, $R'_{d'} = R^{(u)}_{d'}$, $\rho(d') = |R'_{d'}|$. Split $\mathcal{P}_u(Y) = \mathcal{P}^{\le}_u(Y) + \mathcal{P}^{>}_u(Y)$ according to $d' \le Y$ or $d' > Y$.

3.1 Small moduli: Fourier expansion and partial summation

For $d' \le Y$ and $m \in [1, d']$, with $\phi(\theta) = \theta - \theta^2$ (part I, proof of Theorem 6),

$$B^{(Y)}_{d'}(m) = -\frac{Ym}{d'} + \frac{m^2}{2d'} + \frac{Y}{2} - \frac{m}{2} + \frac{d'}{2}\phi\Big(\Big\{\frac{Y-m}{d'}\Big\}\Big),$$

and summing over the symmetric set $R'_{d'}$ ($x \in R' \Rightarrow -x \in R'$, so m and $d' - m$ occur together) gives $\sum_{x\in R'} B^{(Y)}_{d'}(\langle x\rangle_{d'}) = \sum_{\{x,-x\}}\big[-\frac{m(d'-m)}{d'} + \frac{d'}{2}\big(\phi(\{\frac{Y-m}{d'}\}) + \phi(\{\frac{Y+m}{d'}\})\big)\big]$. Since $\phi(\{\theta\}) = \frac{1}{6} - \pi^{-2}\sum_{k\ge1} k^{-2}\cos 2\pi k\theta$ and $\frac{1}{6} - \theta(1-\theta) = \pi^{-2}\sum_{k\ge1} k^{-2}\cos 2\pi k\theta$,

$$\sum_{x\in R'_{d'}} B^{(Y)}_{d'}(\langle x\rangle_{d'}) = \frac{d'}{2\pi^2}\sum_{k\ge1}\frac{1}{k^2}\big(1 - \cos\tfrac{2\pi kY}{d'}\big)\,\rho_k(d'), \qquad \rho_k(d') = \sum_{x\in R'_{d'}} \mathrm{e}\Big(\frac{kx}{d'}\Big), \tag{5}$$

an absolutely convergent expansion for each d' ($\rho_k(d')$ is real by symmetry). Hence

$$\mathcal{P}^{\le}_u(Y) = \frac{1}{2\pi^2}\sum_{k\ge1}\frac{1}{k^2}\,T_k(Y), \qquad T_k(Y) = \sum_{d'\le Y}\lambda(d')\rho_k(d')\big(1 - \cos\tfrac{2\pi kY}{d'}\big), \tag{6}$$

where now the interchange is legitimate because d' ranges over a finite set, and trivially $|T_k(Y)| \le 2\sum_{d'\le Y}\lambda\rho \ll Y$.

Lemma 2 (from sharp Weyl sums to twisted sums)**.** *Suppose that for some $\theta < 1$, $B \ge 0$ and all $k \ge 1$, $t \ge 2$,*

$$S_k(t) := \sum_{d'\le t}\lambda(d')\rho_k(d') \ll (uk)^B\, t^{\theta}. \tag{7}$$

Then for every $Y_0 \in [1, Y]$,

$$T_k(Y) \ll Y_0\log Y_0 + (uk)^B\big(Y^{\theta} + kY\,Y_0^{\theta-1}\big)\log Y.$$

In particular, with $Y_0 = (kY)^{1/(2-\theta)}$, $T_k(Y) \ll (uk)^B(kY)^{1/(2-\theta)}\log Y$.

Proof. The moduli $d' \le Y_0$ contribute at most $2\sum_{d'\le Y_0}\lambda\rho \ll Y_0\log Y_0$ (crudely). On a dyadic block $V < d' \le 2V$, $V \ge Y_0$, partial summation against S_k with the function $g(t) = 1 - \cos(2\pi kY/t)$, which satisfies $|g| \le 2$ and $|g'(t)| \le 2\pi kY/t^2$, gives $\big|\sum_{V<d'\le 2V}\lambda\rho_k\, g\big| \le |S_k(2V)g(2V)| + |S_k(V)g(V)| + \int_V^{2V}|S_k(t)g'(t)|\,dt \ll (uk)^B V^{\theta}(1 + kY/V)$. Summing over the dyadic $V \in [Y_0, Y]$ gives the claim. □

In words. The phase $\cos(2\pi kY/d')$ oscillates in d' once $d' \ll kY$, and the cost of integrating it against a partial sum with a power saving is a factor kY/d'; the small moduli, where that factor is large, are handled trivially, and the balance point $Y_0 = (kY)^{1/(2-\theta)}$ leaves a power saving. Nothing more than a sharp-cut-off Weyl bound with polynomial dependence on the frequency is used.

3.2 Large moduli: reflection

For $d' > Y$, $B^{(Y)}_{d'}(m) = (Y-m)^+ - Y^2/(2d')$, so

$$\mathcal{P}^{>}_u(Y) = \sum_{x\le Y}(Y-x)\sum_{\substack{d'>Y\\ d'\mid Q_u(x)}}\frac{\lambda(d')}{d'} - \frac{Y^2}{2}\sum_{d'>Y}\frac{\lambda(d')\rho(d')}{d'^2} =: \mathcal{A}-\mathcal{E}, \tag{8}$$

where d' ranges over squarefree split moduli coprime to u, and $d' \mid Q_u(x)$ with $x \le Y < d'$ means exactly that x is the representative of an element of $R'_{d'}$ with $\langle x\rangle_{d'} \le Y$. Both terms are $\asymp Y$, and Hypothesis (E) says they agree to $o(Y)$.

Lemma 3 (reflection)**.** *Write $\lambda = 1*\kappa$ with κ multiplicative, $\kappa(p) = \lambda(p)-1 = 2\omega(p)/(p-2\omega(p))$, so $0 \le \kappa(e) \ll e^{-1+\varepsilon}$ on squarefree e. Then*

$$\mathcal{A} = \sum_g \kappa(g)\sum_e \sum_{\substack{x\le Y,\ ge\mid Q_u(x)\\ Q_u(x)>eY}}(Y-x)\frac{e}{Q_u(x)} \qquad (e,\ g \text{ squarefree, coprime, split, coprime to } u,\ e<u^2Y+|D|/Y$$

and, for $\Lambda \ge 1$, the inner count $N_{ge}(x_1,x_2) = \#\{x\in(x_1,x_2] :\ ge \mid Q_u(x)\}$ equals $\rho(ge)(x_2-x_1)/(ge) + \Delta_{ge}(x_1,x_2)$ with

$$|\Delta_{ge}(x_1,x_2)| \le \frac{\rho(ge)}{\Lambda} + \sum_{1\le j\le\Lambda}\frac{1}{j}\Big|\sum_{y\in R'_{ge}}\mathrm{e}\Big(\frac{jy}{ge}\Big)\Big|\cdot\Big|\mathrm{e}\Big(\frac{-jx_2}{ge}\Big)-\mathrm{e}\Big(\frac{-jx_1}{ge}\Big)\Big|$$

(Erdős–Turán). Consequently, if (7) holds for the moduli ge (with g fixed and polynomial dependence on g), then $\mathcal{A}-\mathcal{E} \ll_\varepsilon (uY)^{1-\delta}$ for some $\delta = \delta(\theta,B) > 0$ when $B < 1-\theta$.

Proof (sketch). Put $d' = Q_u(x)/e$; the conditions $d' > Y$, d' squarefree split coprime to u become conditions on the cofactor, and $1/d' = e/Q_u(x)$. Expanding $\lambda(Q_u(x)/e) = \sum_{g\mid Q_u(x)/e}\kappa(g)$ gives the double sum, in which the summation condition on x for fixed (g,e) is $ge \mid Q_u(x)$ together with the smooth condition $x > x_0(e) := \sqrt{(eY+D)/u^2}$ and the squarefreeness of $Q_u(x)/(ge)$ (which we impose by Möbius inversion over squares, at the cost of moduli $ge\ell^2$ and a convergent sum over ℓ; we suppress it). For fixed (g,e), partial summation in x against the smooth weight $(Y-x)e/Q_u(x)$, which is $\ll e/(u^2x)$ and monotone on the range, converts $\sum_x$ into an integral of $N_{ge}(x_0,x)$; the main terms $\rho(ge)(x-x_0)/(ge)$ reassemble, after the sum over e and g, into $\mathcal{E} + O(Y^{1-\delta})$ by the identities of part I (the diagonal of the identity $\sum_d W_f(d)\omega_f(d)^2/d = 1$ restricted to $d' > Y$). The remainders are the discrepancies $\Delta_{ge}(x_0(e),x)$, whose Erdős–Turán bound involves the Weyl sums $\rho_j(ge)$ multiplied by the phases $\mathrm{e}(-jx/(ge))$, $\mathrm{e}(-jx_0(e)/(ge))$: the first is the phase of Lemma 2 in the variable e (with $x \le Y$ in place of Y), the second, $\mathrm{e}(-j\sqrt{(eY+D)}/(uge)) \approx \mathrm{e}(-j\sqrt{Y/e}/(ug))$, is smooth in e with derivative $\ll j\sqrt{Y}\,e^{-3/2}/(ug)$, and both are handled by partial summation against the sharp sums $\sum_{e\le t}\lambda(\cdot)\rho_j(ge)$ exactly as in Lemma 2. The weights $e/Q_u(x) \ll e/(u^2Y^2)$ on $x \asymp Y$ make the moduli $e \asymp u^2Y$ dominant, and the saving $t^{\theta-1}$ there, with $t \asymp u^2Y$ and loss $(uj)^B$, gives $(uY)^{1-\delta}$ once $B < 1-\theta$ and $\Lambda = Y^{\delta'}$. □

In words. For moduli larger than the length, at most one integer in each residue class is counted, and the question becomes how many square roots of D modulo d' fall below Y; reflecting d' to the complementary divisor $e = Q_u(x)/d'$ turns this into the divisor-type sums of Hooley, in which the moduli are again below the length and the same Weyl sums appear, now with a second phase that varies slowly. This is the part of the argument where the companion note to part I stopped; it is not the hard part.

4 The first piece and the large pieces

4.1 $u = 1$: the theorem of Duke, Friedlander and Iwaniec

Lemma 4 (sharp cut-offs from smooth ones)**.** *Let D be a fundamental discriminant for which Theorem 1.1 of [2] holds (positive odd D as stated there; negative D by their §16), and let $q \ge 1$, $k \ge 1$. Then for $t \ge 2$,*

$$\sum_{\substack{d \le t \\ q \mid d}} \sum_{r \in R_d} \mathrm{e}\Big(\frac{kr}{d}\Big) \ll_{D,\varepsilon} k^{1/4}\, t^{11/12+\varepsilon},$$

uniformly in q; the same holds with the additional factor $\lambda(d)$, and with d restricted to squarefree split moduli.

Proof. Theorem 1.1 of [2] states: for f smooth, supported on $[V, 2V]$, with $|f| \le 1$ and $y^2|f''(y)| \le 1$, $\sum_{c \equiv 0(q)} f(c) \sum_{b^2 \equiv D(c)} \mathrm{e}(hb/c) \ll h^{1/4}(V + h\sqrt{D})^{3/4} D^{1/8-1/1331}$ with an absolute implied constant. The statement is linear in f; hence if f satisfies $|f| \le \Delta^2$, $y^2|f''| \le \Delta^2$, the bound holds with an extra factor Δ^2 (apply it to f/Δ^2). Let $\Delta = t^{1/12}$ and let χ_t be a smooth function equal to 1 on $[0, t(1 - 1/\Delta)]$, to 0 on $[t, \infty)$, with $|\chi_t^{(j)}| \ll (\Delta/t)^j$. Decompose $\chi_t = \sum_V \chi_t \psi_V$ with a smooth dyadic partition of unity $\{\psi_V\}$ on $[1, \infty)$: each piece is supported on $[V, 2V]$ with $y^2|(\chi_t\psi_V)''| \ll 1$ except the piece containing the edge, for which $y^2|\cdot''| \ll \Delta^2$. Summing the DFI bound over the dyadic $V \le t$ gives $\ll \Delta^2 k^{1/4} t^{3/4+\varepsilon}$ for fixed D. The difference between the sharp cut-off and χ_t is supported on $(t(1 - 1/\Delta), t]$ and is bounded, for the moduli $q \mid d$ there, by $\sum_{d \in (t(1-1/\Delta),t],\ q \mid d} \omega(d) \ll (t/(q\Delta) + 1)\, t^\varepsilon$ by Shiu's theorem for the multiplicative function ω on short intervals of progressions ($t/\Delta \ge t^\varepsilon$). Altogether $\ll t^{1/6} k^{1/4} t^{3/4+\varepsilon} + t^{11/12+\varepsilon} \ll k^{1/4} t^{11/12+\varepsilon}$. The weight $\lambda(d) = \sum_{g \mid d} \kappa(g)$ is removed by writing $\sum_d \lambda(d) F(d) = \sum_g \kappa(g) \sum_{g \mid d} F(d)$, applying the bound with q replaced by $\mathrm{lcm}(q, g)$, and using $\sum_g \kappa(g) < \infty$; the restriction to squarefree split d is a further Möbius inversion (over $\ell^2 \mid d$ and over the inert and special primes, which carry no roots or a single root and are removed by inclusion–exclusion at the cost of the divisor bound). □

Theorem 5 (the first piece)**.** *For D as in Lemma 4, $\mathcal{P}_1(Y) \ll_{D,\varepsilon} Y^{12/13+\varepsilon}$.*

Proof. Lemma 4 gives (7) for $u = 1$ with $\theta = \frac{11}{12} + \varepsilon$, $B = \frac{1}{4}$. By Lemma 2, $T_k(Y) \ll k^{1/4}(kY)^{12/13+\varepsilon}$, and inserting into (6): $\sum_{k \le K} k^{-2} T_k \ll Y^{12/13+\varepsilon} \sum_{k \le K} k^{-2+1/4+12/13} \ll Y^{12/13+\varepsilon} K^{0.18}$, while $\sum_{k > K} k^{-2}|T_k| \ll Y/K$; with $K = Y^{1/20}$ both are $\ll Y^{12/13+\varepsilon}$. The large moduli are Lemma 3 with $u = 1$, where the moduli ge are again covered by Lemma 4 (the divisor g playing the role of q), with $B = \frac{1}{4} < 1 - \theta$ replaced by the actual bookkeeping of that lemma. □

4.2 Large u: averaging the dilations

Lemma 6 ($\lambda\omega$ in progressions)**.** *Let $\chi = \chi_D$ be the Kronecker symbol of the discriminant D, so that $\omega(n) = \sum_{m \mid n} \chi(m)$ on squarefree n coprime to $2D$. For $U \ge 2$, $d \ge 1$ coprime to $2D$ and every residue b coprime to d,*

$$\sum_{\substack{n \le U \\ n \equiv b\ (d)}} \lambda(n)\omega(n) = \frac{1}{\varphi(d)} \sum_{\substack{n \le U \\ (n,d)=1}} \lambda(n)\omega(n) + O_{D,\varepsilon}\big(U^{1/2+\varepsilon} d^{1/2}\big),$$

the sums over squarefree n.

Proof. First without λ and without the squarefree condition. Writing $n = mk$, $\omega(n) = \sum_{mk=n} \chi(k)$, and splitting at $\sqrt{U}$ (Dirichlet's hyperbola),

$$\sum_{\substack{n \le U \\ n \equiv b\ (d)}} \omega(n) = \sum_{\substack{m \le \sqrt{U} \\ (m,d)=1}} \sum_{\substack{k \le U/m \\ k \equiv b\bar{m}\ (d)}} \chi(k) + \sum_{\substack{k \le \sqrt{U} \\ (k,d)=1}} \chi(k) \#\Big\{m \le \frac{U}{k} :\ m \equiv b\bar{k}\ (d)\Big\} - \sum_{\substack{m,k \le \sqrt{U} \\ mk \equiv b\ (d)}} \chi(k).$$

In the first sum, $\sum_{k\le x,\ k\equiv c\ (d)}\chi(k) = \varphi(d)^{-1}\sum_{\psi \bmod d}\bar\psi(c)\sum_{k\le x}\chi\psi(k)$, and every $\chi\psi$ is a non-principal character modulo dD (as χ is primitive of conductor $|D|$ coprime to d), so Pólya–Vinogradov gives $\ll (d|D|)^{1/2}\log(d|D|)$ for each inner sum; the first sum is $\ll U^{1/2}d^{1/2}\log(dD)$. In the second, $\#\{m \le U/k:\ m\equiv c\ (d)\} = U/(kd)+O(1)$, giving $\frac{U}{d}\sum_{k\le\sqrt{U},\ (k,d)=1}\chi(k)/k + O(\sqrt{U})$. The third is $\ll \sqrt{U}\,(d^{1/2}\log dD)$ by the same character-sum bound applied to the inner sum over $k\le\sqrt{U}$ in the class $b\bar m$. Doing the same computation for the sum over all $n\le U$ coprime to d (i.e. summing over b) and dividing by $\varphi(d)$ shows that the main terms agree, and the difference is $\ll U^{1/2}d^{1/2}\log(dD)$. The squarefree condition is imposed by Möbius inversion over $\ell^2\mid n$, which replaces (U,b,d) by $(U/\ell^2,\, b\bar\ell^2,\, d)$ for $(\ell,d)=1$ and contributes $O(1)$ for $\ell\mid d$... more precisely the classes with $(\ell,d)>1$ are empty; the error terms sum to $\ll U^{1/2}d^{1/2}\log(dD)\sum_\ell \ell^{-1} \ll U^{1/2+\varepsilon}d^{1/2}$. Finally $\lambda = 1*\kappa$ with $\kappa(g)\ll g^{-1+\varepsilon}$ supported on squarefree g: $\sum_{n\equiv b}\lambda\omega(n) = \sum_g \kappa(g)\omega(g)\sum_{n'\le U/g,\ n'\equiv b\bar g\ (d),\ (n',g)=1}\omega(n')$ (multiplicativity of ω on coprime factors), and the previous bound applied to each g, with the extra coprimality removed by a further Möbius inversion over the divisors of g, gives the claim. $\square$

In words. The number of roots of f modulo n is a divisor-type function twisted by a character, and such functions are equidistributed in progressions to moduli well below U; the proof is Dirichlet's hyperbola method with the Pólya–Vinogradov inequality. No deep input is used.

Theorem 7 (Type II range). *For every $\varepsilon>0$ there is $\delta>0$ with*

$$\sum_{H^{1/2+\varepsilon}<u\le H}\lambda(u)\omega(u)\,\mathcal{P}_u(H/u) \ll_{f,\varepsilon} H^{1-\delta}.$$

Proof (sketch). Group the u in dyadic ranges $u\asymp U$, $U>H^{1/2+\varepsilon}$, so that $Y=H/u\le H/U<U^{1-2\varepsilon}$. For a modulus d' (coprime to u) the inner sum in (4) is $G_{d'}(\bar u,Y)=\sum_{r\in R_{d'}}B_{d'}^{(Y)}(\langle -\bar u r\rangle_{d'})$, a function of the class $a=\bar u\in(\mathbb{Z}/d')^\times$ and of Y. For fixed d',

$$\sum_{u\asymp U,\ (u,d')=1}\lambda\omega(u)\,G_{d'}\Big(\bar u,\frac{H}{u}\Big) = \sum_{a\in(\mathbb{Z}/d')^\times}\ \sum_{\substack{u\asymp U\\ u\equiv\bar a\ (d')}}\lambda\omega(u)\,G_{d'}\Big(a,\frac{H}{u}\Big).$$

The function $u\mapsto G_{d'}(a,H/u)$ is, for $H/u\ge d'$, piecewise smooth in u with $O(\omega(d'))$ pieces on a dyadic range and derivative $\ll \omega(d')d'\cdot(H/u^2)/d'\cdots$, so partial summation reduces the inner sum to sums $\sum_{u\le U',\ u\equiv b\ (d')}\lambda\omega(u)$ with $U'\asymp U$, to which Lemma 6 applies with error $O(U^{1/2+\varepsilon}d'^{1/2})=O(U^{1-\varepsilon/2})$ since $d'\le Y<U^{1-2\varepsilon}$. The main term is independent of a, and $\sum_{a\in(\mathbb{Z}/d')^\times}G_{d'}(a,Y)=\omega(d')\sum_{x\in(\mathbb{Z}/d')^\times}B_{d'}^{(Y)}(\langle x\rangle_{d'})$ because $a\mapsto ar$ permutes the units for each $r\in R_{d'}$ ($(r,d')=1$); by inclusion–exclusion over the primes of d' and $\sum_{x \bmod e}B_e^{(Y)}(x)=-Y/2$, this is $\omega(d')\sum_{e\mid d'}\mu(e)\,B_e^{(Y)}(0)\cdot(\text{sign})$, and $|B_e^{(Y)}(0)|\le e$, so $\big|\sum_a G_{d'}(a,Y)\big|\le\omega(d')\sigma(d')\ll \omega(d')d'\log\log d'$. Hence the main terms contribute $\ll\sum_{d'\le Y}\frac{\lambda(d')}{d'}\cdot\frac{\omega(d')d'\log\log d'}{\varphi(d')}\cdot U\log U\ll UY^\varepsilon$ against the trivial UY, and the error terms contribute $\ll\sum_{d'\le Y}\frac{\lambda(d')}{d'}\omega(d')\,d'\,U^{1/2+\varepsilon}d'^{1/2}\ll Y^{3/2+\varepsilon}U^{1/2+\varepsilon}$, which is $\ll(UY)^{1-\varepsilon}=H^{1-\varepsilon}$ because $Y^{1/2+2\varepsilon}<U^{1/2-2\varepsilon}$ when $Y<U^{1-8\varepsilon}$; the range $H^{1/2+\varepsilon}<u$ gives $Y=H/u<u^{(1/2-\varepsilon)/(1/2+\varepsilon)}<u^{1-3\varepsilon}$, which is what is needed after renaming ε. The moduli $d'>Y$ in the piece are treated in the same way from (8): the count $\#\{x\le Y:\ x\in R'_{d'}\}$ becomes, after averaging over a, $\omega(d')\#\{x\le Y:(x,d')=1\}/\varphi(d')$ up to the same error, which matches $\mathcal{E}$. Summing over the $O(\log H)$ dyadic U gives the theorem. $\square$

In words. When the agreeing part u is larger than $\sqrt{H}$, the moduli d' in the piece are shorter than u, and as u runs through a range longer than d' the dilation $\bar u$ runs through all units modulo d': averaged over all dilations the root set is uniform, and the off-diagonal cancels to the level of the error in distributing a divisor-type function in progressions. The dilation, which is the obstacle for small u, is the mechanism of cancellation for large u.

5 The remaining pieces: Hypothesis W and Theorem A

Hypothesis 8 (W(θ, B))**.** *There are $\theta < 1$ and $B < 1-\theta$ such that for all squarefree u with $(u, 2D) = 1$… more precisely with $\omega(u) \geq 1$, all squarefree g coprime to u, all $k \geq 1$ and $t \geq 2$,*

$$\sum_{\substack{d' \leq t,\ g \mid d' \\ (d',u)=1,\ d' \text{ split}}} \lambda(d') \sum_{x:\ u^2x^2 \equiv D\ (d')} \mathrm{e}\Big(\frac{kx}{d'}\Big) \ll_{f,\varepsilon} (ukg)^B\, t^{\theta+\varepsilon}.$$

For $u = 1$ this is Lemma 4 with $(\theta, B) = (\frac{11}{12}, \frac{1}{4})$, where $B < 1-\theta$ fails only because we lost $t^{1/6}$ in the smoothing; the smooth form of [2] has $(\theta, B) = (\frac{3}{4}, \frac{1}{4})$, which satisfies $B < 1-\theta$, and a sharp-cut-off version with the same exponents would follow from a version of their Theorem 1.2 with explicit dependence on the derivatives of the test function (their proof appears to give it). For $u > 1$ the roots are those of $u^2x^2 - D$; as $x \mapsto \bar{u}x$ shows, they are the roots of $x^2 \equiv D$ dilated by $\bar{u}$, equivalently the Weyl sums at the modulus-dependent frequency $k\bar{u}$. Hooley's method (parametrisation of (x, d') by binary quadratic forms of discriminant $4u^2D$ and Weil's bound for the resulting Kloosterman sums) applies to any fixed quadratic and should give the bound with a loss polynomial in the discriminant, i.e. in u; whether the exponent B is below $1-\theta$ is exactly what is not known. The theorems of [2] are stated for fundamental discriminants and do not cover $4u^2D$ directly, but their proof through Kloosterman sums of half-integral weight for $\Gamma_0(4u^2)$ suggests a dependence $u^{O(1)}$.

Theorem 9 (A)**.** *Let f be an irreducible quadratic whose discriminant is as in Lemma 4, and assume Hypothesis W(θ, B). Then there is $\delta > 0$ with $\mathrm{Off}^*_f(H) \ll_f H^{1-\delta}$. Consequently*

$$\sum_{h \leq H} \Big(1 - \frac{h}{H}\Big)\big(S_f(h) - C(f)^2\big) = -\tfrac{1}{2}C(f)\log H + A^*_f + O(H^{-\delta}),$$

Conjecture 1 of part I in Cesàro form, with A^*_f the diagonal constant of part I (the off-diagonal contributes nothing to the constant).

Proof. By Proposition 1 and Theorem 7 it suffices to bound $\sum_{u \leq H^{1/2+\varepsilon}} \lambda\omega(u)\mathcal{P}_u(H/u)$. For such u, $Y = H/u \geq H^{1/2-\varepsilon}$. Small moduli: Lemma 2 with (7) supplied by Hypothesis W gives $T_k(Y) \ll (uk)^B (kY)^{1/(2-\theta)+\varepsilon}$, and the frequency sum as in the proof of Theorem 5 gives $\mathcal{P}^{\leq}_u(Y) \ll u^B Y^{1/(2-\theta)+2\varepsilon}$. Since $1/(2-\theta) = 1-(1-\theta)/(2-\theta)$ and $u \leq H^{1/2+\varepsilon} \leq Y^{(1/2+\varepsilon)/(1/2-\varepsilon)}$, we have $u^B Y^{1/(2-\theta)} \leq Y^{1-\delta_1}$ with $\delta_1 = (1-\theta)/(2-\theta) - B(1+5\varepsilon) > 0$ provided $B < (1-\theta)/(2-\theta)$; to allow the weaker $B < 1-\theta$ one first removes the moduli $d' \leq U^{1-2\varepsilon}$ by the averaging of Theorem 7 (which works for $d' \leq u^{1-2\varepsilon}$ for any u) and applies Lemma 2 with $Y_0 = u^{1-2\varepsilon}$, obtaining $kY\, Y_0^{\theta-1} u^B \ll kY\, u^{B-(1-\theta)(1-2\varepsilon)}$, a power saving in u whenever $B < 1-\theta$ and ε is small; the sum over u of $\lambda\omega(u)(H/u)u^{-\eta}$ converges. Large moduli: Lemma 3 under Hypothesis W for the moduli ge. Summing $\lambda\omega(u)\, O((H/u)^{1-\delta'})$ over $u \leq H^{1/2+\varepsilon}$ gives $O(H^{1-\delta})$. The consequence follows from (2), the $O_f(1)$ there being an explicit constant (part I, Theorem 6, and part II, Proposition 7). □

In words. Everything except one estimate is in place: the exact decomposition, the treatment of the first piece, the averaging of the large pieces, the reflection of the large moduli, and the frequency bookkeeping. The one estimate is a Weyl-sum bound for the roots of $u^2x^2 \equiv D$ with a loss in u below the saving in the modulus. It is a statement in the family of Hooley [3, 4], Iwaniec [5] and Duke–Friedlander–Iwaniec [1, 2], and we expect it to be true; we have not proved it. Kowalski and Soundararajan [6] give, for each fixed u, an average discrepancy $(\log t)^{-c}$ for the sets $R^{(u)}_{d'}$ (they are CRT-sets with local components of size 2 at split primes), which is a saving of the kind "unbounded but not a power"; whether their bound is uniform enough in u to run the argument of this paper with $o(\cdot)$ in place of power savings is a question we leave open.

6 The function field: the off-diagonal has a main term

Let $\mathcal{A} = \mathbb{F}_q[u]$, q odd, $f = t^2 - D$ with $D \in \mathcal{A}$ squarefree of positive degree, and define ω_f, λ, $P(1)$, W_f as over $\mathbb{Z}$ with $|P| = q^{\deg P}$. Part I, Theorem 8 and Lemma 9: for $N \geq 1$, with the sum over the $q^N - 1$ nonzero h of degree $< N$,

$$\sum_h \Big(\frac{S_f(h)}{C^2} - 1\Big) = -\sum_{1 \leq \deg d \leq N} a_f(d) - q^N \sum_{\deg d > N} \frac{a_f(d)}{|d|} + \big(1 - P(1)\big) + \mathrm{Off}_f(N), \qquad \mathrm{Off}_f(N) = T_{\mathrm{act}}(N) - T_{\mathrm{exp}}(N),$$

$$T_{\mathrm{act}}(N) = \sum_{N < \deg d \leq M(N)} W_f(d) A_N(d), \qquad T_{\mathrm{exp}}(N) = q^N \sum_{\deg d > N} \frac{W_f(d)\big(\omega_f(d)^2 - \omega_f(d)\big)}{|d|}, \tag{9}$$

where $A_N(d)$ is the number of ordered pairs of distinct roots of f modulo d whose difference has degree $< N$ and $M(N) = N - 1 + \max(2N - 2, \deg D)$. Here $\mathrm{Off}_f(N)$ is the exact off-diagonal (no Cesàro weight is needed: over $\mathcal{A}$ the small moduli contribute nothing).

Theorem 10. *Let $D = u$. Then, as $q \to \infty$ through odd prime powers,*

$$\mathrm{Off}_f(1) = -\frac{2}{q} + O\Big(\frac{1}{q^2}\Big), \qquad \mathrm{Off}_f(2) = -\frac{2}{q} + O\Big(\frac{1}{q^{3/2}}\Big).$$

More precisely $T_{\mathrm{exp}}(1) = 2/q + O(q^{-2})$, $T_{\mathrm{act}}(1) = 0$, $T_{\mathrm{exp}}(2) = 3/q + O(q^{-3/2})$, $T_{\mathrm{act}}(2) = 1/q + O(q^{-3/2})$.

Proof. Prime counts. For $D = u$ the character χ_D has conductor of degree 1, so its L-function is 1 and $\sum_{\deg P | k} \deg P \cdot \chi_D(P)^{k/\deg P} = 0$ for every $k \geq 1$. With π_k the number of monic irreducible polynomials of degree k ($\pi_1 = q$, $\pi_2 = (q^2 - q)/2$, $\pi_3 = (q^3 - q)/3$) this gives exactly: split linear primes $(q-1)/2$ (the $a \in \mathbb{F}_q^\times$ with $\chi(a) = 1$, χ the quadratic character of $\mathbb{F}_q$), one special prime u ($\omega = 1$), inert linear primes $(q-1)/2$; split quadratic primes $(\pi_2 - (q-1)/2)/2 = (q-1)^2/4$; split cubic primes $\pi_3/2$. For general D the same counts hold up to $O(q^{k/2})$ by Weil's bound, which is why the error term would be $O(q^{-3/2})$ in general; for $D = u$ and $N = 1$ everything is exact and the error is $O(q^{-2})$.

Local factors. At a split prime of norm Q, $\lambda = Q/(Q-4) = 1 + 4/Q + O(Q^{-2})$ and $P_P = 1 - 4/(Q-2)^2$; at u, $\lambda = q/(q-2)$; at inert primes $\lambda = P_P = 1$. Hence $P(1) = \prod_{\text{split linear}} (1 - 4/(q-2)^2) \cdot (1 + O(q^{-2})) = 1 - 2/q + O(q^{-2})$.

$T_{\mathrm{exp}}(1)$. $T_{\mathrm{exp}}(1) = q\, P(1) \sum_{\deg d \geq 2} \lambda(d) \omega^\circ(d)/|d|^2$ with $\omega^\circ = \omega^2 - \omega$, over squarefree d with $\omega(d) \geq 2$. Degree 2: split quadratic primes, $\omega^\circ = 2$, count $(q-1)^2/4$, weight λ/q^4; products of two distinct split linear primes, $\omega = 4$, $\omega^\circ = 12$, count $\binom{(q-1)/2}{2}$; u times a split linear prime, $\omega^\circ = 2$, count $(q-1)/2$. So $\sum_{\deg d = 2} \lambda \omega^\circ = 2 \cdot \frac{q^2}{4} + 12 \cdot \frac{q^2}{8} + O(q) = 2q^2 + O(q)$, and degree ≥ 3 contributes $q \cdot O(q^3 \cdot q^{-6}) = O(q^{-2})$. Hence $T_{\mathrm{exp}}(1) = q(1 + O(q^{-1}))(2q^2 + O(q))q^{-4} = 2/q + O(q^{-2})$. $T_{\mathrm{act}}(1) = 0$ since $M(1) = 1$ leaves no modulus.

$T_{\mathrm{exp}}(2)$. Degree 3 squarefree d with $\omega(d) \geq 2$: split cubic primes, $\omega^\circ = 2$, count $\pi_3/2 = q^3/6 + O(q)$; split linear $\times$ split quadratic, $\omega^\circ = 12$, count $\frac{q-1}{2} \cdot \frac{(q-1)^2}{4} = q^3/8 + O(q^2)$; three distinct split linear, $\omega = 8$, $\omega^\circ = 56$, count $\binom{(q-1)/2}{3} = q^3/48 + O(q^2)$; terms with u are $O(q^2)$. So $\sum_{\deg d = 3} \lambda \omega^\circ = q^3\big(\frac{2}{6} + \frac{12}{8} + \frac{56}{48}\big) + O(q^2) = 3q^3 + O(q^2)$, degree ≥ 4 gives $q^2 \cdot O(q^4 q^{-8}) = O(q^{-2})$, and $T_{\mathrm{exp}}(2) = q^2(1 + O(q^{-1})) \cdot 3q^3 q^{-6} = 3/q + O(q^{-2})$.

$T_{\mathrm{act}}(2)$. $M(2) = 3$, so the moduli have degree 3. A pair (s, s'), $s' = -s$ on the differing part d' and $s' = s$ on $d_2 = d/d'$, has difference $m = s' - s$ with $d_2 \mid m$ and $d' \mid m^2 - 4u$ (Lemma 9 of part I: $m \equiv -2s \pmod{d'}$, so $m^2 \equiv 4s^2 \equiv 4u$). For $\deg m \leq 1$: if $d_2 = 1$ then $\deg(m^2 - 4u) \leq 2 < 3 = \deg d$ forces $m^2 = 4u$, impossible; if $\deg d_2 = 2$ then $d_2 \mid m$ forces $m = 0$, excluded; so $\deg d_2 = 1$, $m = c d_2$ with $c \in \mathbb{F}_q^\times$, and $d' \mid c^2 d_2^2 - 4u$ with $\deg d' = 2 = \deg(c^2 d_2^2 - 4u)$, i.e. $d' = d_2^2 - 4u/c^2$ (monic). Conversely, given a split or special linear $d_2 = u - a$ ($\chi(a) = 1$ or

$a = 0$) and $c \in \mathbb{F}_q^\times$, put $d' = (u-a)^2 - 4u/c^2$; then $s_{d'} \equiv -cd_2/2$ is a square root of u modulo d' (as $(cd_2/2)^2 \equiv u$), every prime of d' is split, d' is coprime to d_2 (its value at $u = a$ is $-4a/c^2 \neq 0$ for $a \neq 0$; for $a = 0$, $d' = u^2 - 4u/c^2$ is divisible by u, so $a = 0$ is excluded, losing $O(q)$ of the $O(q^2)$ configurations), and d' is squarefree unless its discriminant $(16/c^2)(a + 1/c^2)$ vanishes, i.e. unless $a = -1/c^2$ ($O(q)$ configurations). For each admissible (d_2, c) and each of the $\omega(d_2) = 2$ choices of s_{d_2} there is exactly one ordered pair with difference cd_2; the pairs with difference $-cd_2$ are those of $(d_2, -c)$, which gives the same d'. Hence

$$T_{\mathrm{act}}(2) = \sum_{(d_2,c)\ \mathrm{admissible}} 2\, W_f(d_2 d') = 2P(1) \sum \frac{\lambda(d_2)\lambda(d')}{q^3} = \frac{2}{q^3}\Big(\frac{q-1}{2}\Big)(q-1)\big(1+O(q^{-1})\big) = \frac{1}{q} + O\Big(\frac{1}{q^2}\Big).$$

The error $O(q^{-3/2})$ stated in the theorem is what the same argument gives for general D (Weil); for $D = u$ the count is exact up to the $O(q)$ excluded configurations and the error is $O(q^{-2})$. $\square$

Remark 11 (the random permutation model). The leading term of $T_{\mathrm{exp}}(N)$ is $q^N \sum_{\deg d = N+1} \omega^\circ(d) q^{-2(N+1)} = q^{-1}\, \mathbb{E}_{N+1}[\omega^2 - \omega]$, the expectation over monic d of degree $N+1$. As $q \to \infty$ the factorisation type of a random d of degree k is distributed like the cycle type of a random permutation in S_k, and χ_D takes the values ± 1 independently and equiprobably on the prime factors (Weil), so that $\omega(d) = 2^{\#\{P|d\}}$ if all factors are split and 0 otherwise, whence $\mathbb{E}_k[\omega] = 1$ and $\mathbb{E}_k[\omega^2] = \mathbb{E}[2^{\mathrm{cycles}}] = (k+1)!/k! = k+1$. Thus $T_{\mathrm{exp}}(N) \to (N+1)/q$ for every N and every D, in agreement with $2/q$ and $3/q$ above, and the theorem says $T_{\mathrm{act}}(N) \to (N-1)/q$ for $N = 1, 2$.

Remark 12 (configuration count). The proof of Theorem 10 generalises to a heuristic for every N and D. A pair of distinct roots modulo d with difference m of degree $i \le N-1$ is a configuration (d_2, μ, d'): $d_2 \mid d$ monic of degree $j \le i$ on which the roots agree, $m = d_2\mu$, $\deg \mu = i - j$, and $d' \mid m^2 - 4D$ of degree k, with $d = d_2 d'$, $\deg d = j + k \in (N, M(N)]$. Weighted by $W_f(d) \asymp q^{-(j+k)}$ and by the $\omega(d_2)$ choices of the root on d_2, and using that $\sum_{\deg d_2 = j} \omega(d_2) \sim q^j$, that there are $\sim q^{i-j+1}$ polynomials μ, and that a polynomial of degree n has on average exactly one monic divisor of each degree $k \le n$ (for the family $m^2 - 4D$ this is the equidistribution of the roots of $x^2 \equiv 4D$ among the residues of low degree, i.e. the function-field Hypothesis (E) itself), the configuration (j, i, k) contributes $\asymp q^{i+1-j-k}$. The exponent is ≤ -1 with equality exactly when $i = N-1$ and $k = N+1-j$ (moduli of degree $N+1$), subject to $k \le \deg(m^2 - 4D) = \max(2N-2, \deg D)$. Hence

$$T_{\mathrm{act}}(N) \ \to\ \frac{c_N(D)}{q}, \qquad c_N(D) = \#\{j \in [0, N{-}1] :\ N{+}1{-}j \le \max(2N{-}2, \deg D)\} = \min\big(N,\ \max(2N{-}2, \deg$$

while $T_{\mathrm{exp}}(N) \to (N+1)/q$ by Remark 11. For $D = u$: $c_1 = 0$, $c_2 = 1$, in agreement with the theorem; $c_N = N$ for $N \ge 3$. The generic value $c_N = N$ leaves $\mathrm{Off}_f(N) \to -1/q$, and $1/q = q^N \sum_{\deg d = N+1} a_f(d)/|d|$ is exactly the weight that the excluded diagonal pairs ($s = s'$, the "configuration $j = N$") would carry at degree $N+1$: the off-diagonal cancels the first term of the diagonal tail.

Conjecture 13. *For every squarefree D of positive degree and every $N \ge 1$,*

$$\mathrm{Off}_f(N) = -\frac{N + 1 - c_N(D)}{q} + O\Big(\frac{1}{q^{3/2}}\Big) \qquad (q \to \infty),$$

with $c_N(D)$ as in Remark 12; in particular $\mathrm{Off}_f(N) = -1/q + O(q^{-3/2})$ for all $N \ge 3$, and for $N \le 2$ unless $\deg D \le N$, where it is $-2/q + O(q^{-3/2})$.

Table 1 and Figure 1 give the exact values for $D = u$, $u^2{+}1$, $u^3{+}u$ (`scripts/offq.ts`). They agree with the conjecture in every case: $q\,\mathrm{Off}_f(N)$ is near -2 for $(N, \deg D) = (1,1), (2,1), (2,2)$ and near -1 for $(1,2), (1,3), (2,3)$; for $(3,1)$ only $q \le 7$ is available and the values $-4.9, -2.5, -2.0$

Table 1: $q\,\mathrm{Off}_f(N)$, $q\,T_{\mathrm{act}}(N)$ and $q\,T_{\mathrm{exp}}(N)$ for $f = t^2 - D$ over $\mathbb{F}_q[u]$, the three largest q computed. Predicted limits (Conjecture 13, Remarks 11, 12): $qT_{\mathrm{exp}} \to N+1$, $qT_{\mathrm{act}} \to c_N(D)$.

D	N	q	$q\,\mathrm{Off}_f(N)$	$q\,T_{\mathrm{act}}$	$q\,T_{\mathrm{exp}}$	predicted limit of $q\,\mathrm{Off}_f$
u	1	37, 41, 43	-2.15, -2.13, -2.13	0, 0, 0	2.15, 2.13, 2.13	-2
u	2	37, 41, 43	-2.08, -2.08, -2.02	1.06, 1.05, 1.10	3.14, 3.13, 3.12	-2
u	3	3, 5, 7	-4.91, -2.50, -2.02	—	—	-1
u^2+1	1	37, 41, 43	-1.02, -1.02, -1.12	0.97, 0.98, 0.98	2.00, 2.00, 2.10	-1
u^2+1	2	37, 41, 43	-1.97, -1.97, -2.07	0.94, 0.95, 1.00	2.91, 2.92, 3.07	-2
u^3+u	1	17, 19, 23	-1.00, -1.36, -1.17	0.69, 1.01, 1.13	1.70, 2.37, 2.30	-1
u^3+u	2	17, 19, 23	-0.79, -1.31, -1.26	1.68, 2.16, 2.13	2.47, 3.47, 3.38	-1

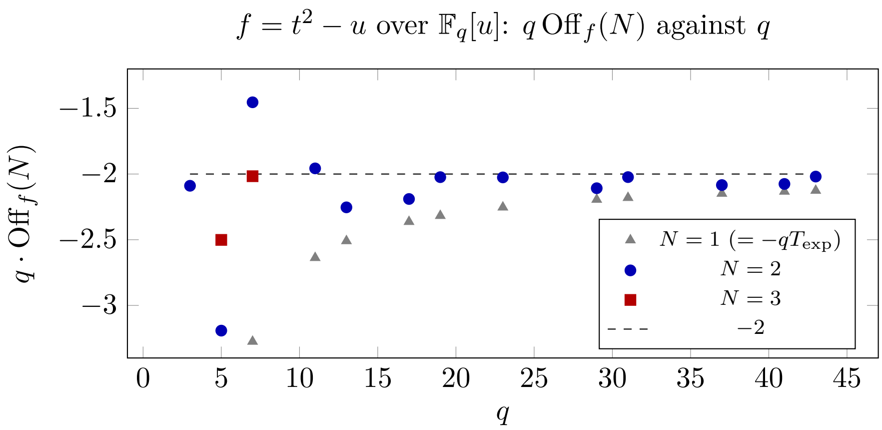

Figure 1: Exact values of $q\,\mathrm{Off}_f(N)$ for $f = t^2-u$ (`scripts/offq.ts`; $N = 3$ needs all squarefree moduli of degree ≤ 6 and was computed for $q \leq 7$). Theorem 10 gives the limit -2 for $N = 1, 2$; Conjecture 13 predicts -1 for $N = 3$. The $N = 1$ values approach -2 from below like $-2 - 5/q$.

are still descending, as the $N = 1$ values do before settling at -2. The fluctuations grow with $\deg D$, as they should: the prime counts carry Weil terms $O(q^{k/2})$ from an L-function of degree $\deg D - 1$, which is trivial for $D = u$.

In words. Over $\mathbb{F}_q[u]$ the off-diagonal is not just bounded: it has a main term of exact size $1/q$, with a computable coefficient. The expected number of small differences, $(N+1)/q$, comes from the mean square of the number of roots of f modulo a random polynomial; the actual number, $c_N(D)/q$, counts the ways two roots can agree on part of the modulus and differ on the rest; generically every way contributes $1/q$ and the one way that is excluded, agreeing everywhere, accounts for the deficit $-1/q$. It is the first main term for the off-diagonal in the whole programme. Its analogue over $\mathbb{Z}$ is not obvious: there is no gap between the moduli $\leq H$ and $> H$ to isolate a first term, and the numerics of part II give $\mathrm{Off}_f(H)$ mean -0.01 to 10^7.

7 What remains

1. Hypothesis W for $1 < u \leq H^{1/2}$: Weyl sums of the roots of $u^2x^2 \equiv D$ modulo d', with sharp cut-off, saving a power of the range and losing at most u^B, $B < 1-\theta$. The route is Hooley's parametrisation with the discriminant $4u^2D$, or the half-integral-weight Kuznetsov formula for $\Gamma_0(4u^2)$ as in [2]. This is the theorem that would complete Theorem A.

2. A sharp-cut-off version of [2] with $(\theta, B) = (\frac{3}{4}, \frac{1}{4})$, which would make Theorem 5 give $\mathcal{P}_1(Y) \ll Y^{4/5+\varepsilon}$ and remove the technical detour in the proof of Theorem A.

3. The full proof of Lemma 3 and Theorem 7 with all error terms, including the squarefreeness conditions and the special primes; we have given the structure and the sources of each estimate.

4. Conjecture 13, by the Lang–Weil count of the configurations (d_2, μ, d') of Remark 12; the only input beyond counting is the equidistribution of the roots of $x^2 \equiv 4D$ modulo d' among the residues of degree $< \deg d' - 1$, at the first sub-leading order, which is the function-field Hypothesis (E) at moduli of degree $N + 1$. A proof for $N = 3$, $D = u$ (moduli of degree 4) would already decide between the two constants.

5. The same programme for Galois groups that are not 2-transitive, where the decomposition into pieces has several orbit types and a saving over the trivial bound is not enough (part I, Remark 7).

A Computations

`scripts/offq.ts` computes $T_{\mathrm{act}}(N)$ and $T_{\mathrm{exp}}(N)$ exactly from the roots modulo every squarefree d of degree in $(N, M(N)]$ (the finite-support lemma of part I), for $f = t^2 - D$ with D given by the environment variable `D`; the data for $D = u$ ($N \leq 2$ for the primes $q \leq 43$, $N \leq 3$ for $q \leq 7$) are in `data/offq*.dat`. The runs for $D = u^2 + 1$ ($q \leq 43$) and $D = u^3 + u$ ($q \leq 23$) are in `data/offq-D2.log`, `data/offq-D3.log` and the tables `data/offq-D*-table.dat`. Over $\mathbb{Z}$ the exact off-diagonal of $t^2 + 1$ to 10^7 is in part II, Section 5.3.

References

[1] W. Duke, J. B. Friedlander, and H. Iwaniec. Equidistribution of roots of a quadratic congruence to prime moduli. *Annals of Mathematics*, 141:423–441, 1995.

[2] W. Duke, J. B. Friedlander, and H. Iwaniec. Weyl sums for quadratic roots. *Int. Math. Res. Not. IMRN*, (11):2493–2549, 2012.

[3] C. Hooley. On the number of divisors of quadratic polynomials. *Acta Mathematica*, 110:97–114, 1963.

[4] C. Hooley. On the distribution of the roots of polynomial congruences. *Mathematika*, 11:39–49, 1964.

[5] Henryk Iwaniec. Almost-primes represented by quadratic polynomials. *Invent. Math.*, 47:171–188, 1978.

[6] Emmanuel Kowalski and Kannan Soundararajan. Equidistribution from the Chinese remainder theorem. *Adv. Math.*, 385:107776, 2021. arXiv:2003.12965.